

Take Home Assignment, 1

Multi-Line Dialer (2 lines) + Update CRM (or Lead Management MVP)

Stack: Node.js (Express/Fastify) + React (Vite/Next)

AI tools allowed: Yes (Codex/Copilot/Cursor/Claude/etc.). Show what you verified.

Goal

Build a CRM where an agent runs a **2-line dialer session** over a list of leads. When a call ends, the system **writes a CRM activity record** for that lead (mock CRM).

Submission

Deliverables shall be hosted on any free webhost service, CRM tool and utilize any other free existing platform plans. Email the url, Git repo links to

intern1.aisalesdr@gmail.com and ellee@aisalesdr.co

- Repo with instructions:
 - npm install
 - npm run dev (or separate client/server commands)
- NOTES.md:
 - tradeoffs, what you'd do next
 - how you used AI tools + what you verified

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Part 1- Backend (node.js)

Data Models (it is okay for this to be in memory)

- Lead
 - id, name, company, phone, email
 - crmExternalId (optional)
- Call
 - id, leadId, sessionId, status
 - startedAt, endedAt
 - Call_Status: CONNECTED | NO_ANSWER | BUSY | VOICEMAIL | CANCELED_BY_DIALER
 - providerCallId (string)
- DialerSession
 - id, agentId
 - leadQueue (array of leadIds)
 - concurrency fixed to 2
 - activeCallIds (max 2)
 - winnerCallId (nullable)
 - status: RUNNING | STOPPED
 - metrics: attempted, connected, failed, canceled
- CRMActivity
 - id, leadId, crmExternalId, type (e.g., CALL)
 - callId, disposition, notes, createdAt

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Part 2 – CRM Integration

Implement Mock CRM inside backend

These endpoints represent an external CRM system (store data in memory):

CRM Sync behavior (required)

When a call reaches a terminal outcome:

- If lead has no `crmExternalId`, update contact
- Then create activity with disposition + basic notes
- Save the activity in your own app DB (CRMActivity model) and the mock CRM store.

Idempotency requirement

- The same terminal call event should not create duplicate CRM activities. (Use a simple idempotency key like `callId`.)

API endpoints to inspect CRM results

- GET `/mock-crm/contacts`
- GET `/mock-crm/activities`
- GET `/leads/:id/crm-activities` (your app's view)

Part 3 – Frontend (REACT)

Screen 1: Leads + Session creation

- Display seeded leads list (4–8 leads)
- Select leads (checkboxes)
- “Create Dialer Session” + “Start”

Screen 2: Dialer Session

- Show 2 active “lines” (call cards):
 - lead name, phone, status
- Show session metrics
- Show winner call (if connected)
- Show CRM activity creation status (per call/lead)
- Poll backend every 1–2 seconds

Take Home Assignment 2

Product Refinement Exercise: V2 AI-to-Human Dialer

Objective:

Define a production-ready BPO-style V2 and recommend whether to adopt, extend, integrate, or build.

Flow:

SalesDoc lead → AI navigates gatekeeper → PIC reached → informed human takes over → outcome saved to SalesDoc.

Requirements:

- Call only leads with likely-to-buy signals.
- Show company context to the human before handoff.
- Match call volume to human availability.- AI must not collect PII.
- If no human is available, end safely and mark for callback.

Deliverables:

Product proposal, workflow, solution comparison, prioritized backlog, and 15-minute presentation.

See `V2\_PRODUCT\_REFINEMENT\_EXERCISE.md` for the complete assignment.*